Math 161
Problem set 2 solutions

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1. Assume that the well-ordering principle holds. Suppose that $A \subset \mathbb{N}$ satisfies
   - $1 \in A$ and
   - $n \in A \implies n+1 \in A$.

   We want to show that $A = \mathbb{N}$, so suppose for the sake of contradiction that $A \neq \mathbb{N}$. Let $B := \mathbb{N} \setminus A$, so $B \neq \varnothing$ because $A \neq \mathbb{N}$. By the well-ordering principle B has a least element m. Since $1 \in A$, we have $1 \notin B$ and hence $m \neq 1$. Thus $m - 1 \in \mathbb{N}$, and since m is the least element of B, it follows that $m - 1 \notin B$, i.e. $m - 1 \in A$. But the inductive hypothesis on A implies that $m \in A$, which is a contradiction because $m \in B = \mathbb{N} \setminus A$.

2. First, assume that complete induction holds. Suppose that $A \subset \mathbb{N}$ satisfies the hypotheses of ordinary induction, i.e.
   - $1 \in A$ and
   - $n \in A \implies n+1 \in A$.

   We want to show that $A = \mathbb{N}$. Observe that
   - $1 \in A$ and
   - $1, \cdots, n \in A \implies n \in A \implies n+1 \in A$,

   so $A = \mathbb{N}$ by complete induction.

   Now assume instead that ordinary induction holds. Suppose that $A \subset \mathbb{N}$ satisfies the hypotheses of complete induction, i.e.
   - $1 \in A$, and
   - $1, \cdots, n \in A \implies n+1 \in A$.

   The hypotheses of ordinary induction do not obviously hold for A in this case, so consider instead the set
   $$B := \{n \in \mathbb{N} \mid m \in A \text{ for any } m \leq n\} \subset \mathbb{N}.$$
   Then we have $1 \in B$ because $m \leq 1$ implies $m = 1$, and $1 \in A$. For the inductive hypothesis, note that
   $$n \in B \implies 1, \cdots, n \in A \implies n+1 \in A,$$
   and hence
   $$n \in B \implies 1, \cdots, n+1 \in A \implies n+1 \in B.$$
   Thus $B = \mathbb{N}$ by ordinary induction. On the other hand $B \subset A$ by the definition of B, so we conclude that $A = \mathbb{N}$.

3. (i) Note that
   $$-a - (-b) = -a + b = b - a,$$
   where the equality $-(-b) = b$ was proved in class and the second equality uses commutativity of addition. The hypothesis means that $b - a > 0$, so we obtain $-a - (-b) > 0$, or equivalently $-b < -a$.

(ii) The assumption $a < b$ means that $b - a > 0$. Since positive numbers are closed under multiplication, we obtain $(b-a)c > 0$. But

$$(b-a)c = bc + (-a)c = bc - ac,$$

where we used distributivity and the identity $(-a)c = -ac$ established in class. Thus $bc - ac > 0$, i.e. $ac < bc$.

(iii) We showed in class that $c < 0$ implies $-c > 0$. Since $b - a > 0$ by assumption, we have $(b-a)(-c) > 0$ by the closure of positive numbers under multiplication. But

$$(b-a)(-c) = b(-c) + (-a)(-c) = -bc + ac = ac - bc,$$

where we used distributivity, the identities $b(-c) = -bc$ and $(-a)(-c) = ac$ established in class, and commutativity of addition. Thus $ac - bc > 0$, i.e. $ac > bc$.

(iv) Note that $a > 1 > 0$, and since $a - 1 > 0$ we have $a(a-1) > 0$ by the closure of positive numbers under multiplication. Now

$$a(a-1) = a^2 + a(-1) = a^2 - a,$$

where again we used distributivity and the identity $a(-1) = -a$ shown in class. So $a^2 - a > 0$, i.e. $a^2 > a$.

(v) We have $1 - a > 0$ and $a > 0$, so the closure of positive numbers under multiplication implies that $(1-a)a > 0$. But

$$(1-a)a = a + (-a)a = a - a^2,$$

where we once more used distributivity and the identity $(-a)a = -a^2$ shown in class. This means that $a - a^2 > 0$, i.e. $a^2 < a$.

4. Let $a, b, c, d, e, f \in \mathbb{Z}$ with $b, d, f \neq 0$. Then we have

$$\left(\frac{a}{b} + \frac{c}{d}\right) + \frac{e}{f} = \frac{ad + bc}{bd} + \frac{e}{f} = \frac{(ad+bc)f + (bd)e}{(bd)f}$$

by the definition of addition in $\mathbb{Q}$. On the other hand, we have

$$\frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f}\right) = \frac{a}{b} + \frac{cf + de}{df} = \frac{a(df) + b(cf + de)}{b(df)},$$

again by definition. The denominators of these two fractions are equal by associativity of multiplication: $(bd)f = b(df)$. As for the numerators, we have

$$(ad + bc)f + (bd)e = (ad)f + (bc)f + (bd)e = a(df) + b(cf) + b(de) = a(df) + b(cf + de)$$

by the distributive property, associativity of multiplication, and distributivity again.

5. (i) For the base case $n = 1$, observe that

$$1^2 = 1 = \frac{1 \cdot 2 \cdot 3}{6}.$$

Now suppose that the identity holds for some $n \in \mathbb{N}$, so we have to show that it holds for $n + 1$ to complete the induction. For this, we compute

$$\begin{aligned} 1^2 + \cdots + n^2 + (n+1)^2 &= \frac{n(n+1)(2n+1)}{6} + n^2 + 2n + 1 \\ &= \frac{2n^3 + 3n^2 + n}{6} + \frac{6n^2 + 12n + 6}{6} \\ &= \frac{2n^3 + 9n^2 + 13n + 6}{6}, \end{aligned}$$

where the first step used the inductive hypothesis and the rest was algebra. On the other hand, we have

$$\frac{(n+1)(n+2)(2(n+1)+1)}{6} = \frac{(n^2+3n+2)(2n+3)}{6} = \frac{2n^3+9n^2+13n+6}{6},$$

which establishes the desired identity.

(ii) We proved the identity

$$1+\cdots+n = \frac{n(n+1)}{2}$$

in class. Thus, the identity we are trying to prove can be rewritten as

$$1^3+\cdots+n^3 = \frac{n^2(n+1)^2}{4}.$$

The base case $n=1$ is easy to see from the original identity: $1^3 = 1 = 1^2$. For the inductive step, suppose that the identity holds for some $n \in \mathbb{N}$, and compute

$$\begin{aligned} 1^3+\cdots+n^3+(n+1)^3 &= \frac{n^2(n+1)^2}{4} + n^3+3n^2+3n+1 \\ &= \frac{n^4+2n^3+n^2}{4} + \frac{4n^3+12n^2+12n+4}{4} \\ &= \frac{n^4+6n^3+13n^2+12n+4}{4}, \end{aligned}$$

where used the inductive hypothesis and then simplified. Coming from the other side, we have

$$\frac{(n+1)^2(n+2)^2}{4} = \frac{(n^2+2n+1)(n^2+4n+4)}{4} = \frac{n^4+6n^3+13n^2+12n+4}{4}$$

as needed.

6. (i) Add the two binomial coefficients on the right side of the identity:

$$\begin{aligned} \binom{n}{k-1} + \binom{n}{k} &= \frac{n!}{(k-1)!\cdot(n-k+1)!} + \frac{n!}{k!\cdot(n-k)!} \\ &= \frac{n!\cdot k + n!\cdot(n-k+1)}{k!\cdot(n-k+1)!}. \end{aligned}$$

Here we used the standard technique of making like denominators and then adding the numerators, which works because

$$\frac{a}{c} + \frac{b}{c} = \frac{ac+bc}{c^2} = \frac{a+b}{c}$$

for any $a, b, c \in \mathbb{Z}$ with $c \neq 0$. Now the numerator simplifies to

$$n!\cdot k + n!\cdot(n-k+1) = n!\cdot(n+1) = (n+1)!,$$

which implies that

$$\binom{n}{k-1} + \binom{n}{k} = \frac{(n+1)!}{k!\cdot(n-k+1)!} = \binom{n+1}{k}$$

as desired.

(ii) The base case $n=1$ simply says that

$$(a+b)^1 = a+b = \binom{1}{0}a^0b^1 + \binom{1}{1}a^1b^0.$$

For the inductive step, assume that the binomial formula holds for some $n \in \mathbb{N}$. Then we have

$$\begin{aligned}
(a+b)^{n+1} &= (a+b)^n(a+b) \\
&= \left(\sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}\right)(a+b) \\
&= \sum_{k=0}^{n} \binom{n}{k} (a^{k+1}b^{n-k} + a^k b^{n-k+1})
\end{aligned}$$

Next, we split the sum into two terms and shift the indexing of the first term:

$$\sum_{k=0}^{n} \binom{n}{k} (a^{k+1}b^{n-k} + a^k b^{n-k+1}) = \sum_{k=1}^{n+1} \binom{n}{k-1} a^k b^{n-k+1} + \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k+1}.$$

Separating the terms $k = n+1$ and $k = 0$ from the first and second sum respectively, we can recombine the two sums:

$$\sum_{k=1}^{n+1} \binom{n}{k-1} a^k b^{n-k+1} + \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k+1} = b^{n+1} + \sum_{k=1}^{n+1} \left(\binom{n}{k-1} + \binom{n}{k}\right) a^k b^{n-k+1} + a^{n+1},$$

where we used the fact that

$$\binom{n}{0} = 1 = \binom{n}{n}.$$

Finally, we apply the identity from part (i) and regroup into a single sum to obtain

$$b^{n+1} + \sum_{k=1}^{n+1} \left(\binom{n}{k-1} + \binom{n}{k}\right) a^k b^{n-k+1} + a^{n+1}b^0 = \sum_{k=0}^{n+1} \binom{n+1}{k} a^k b^{n+1-k},$$

as desired. Here we used the fact that

$$\binom{n+1}{0} = 1 = \binom{n+1}{n+1}.$$